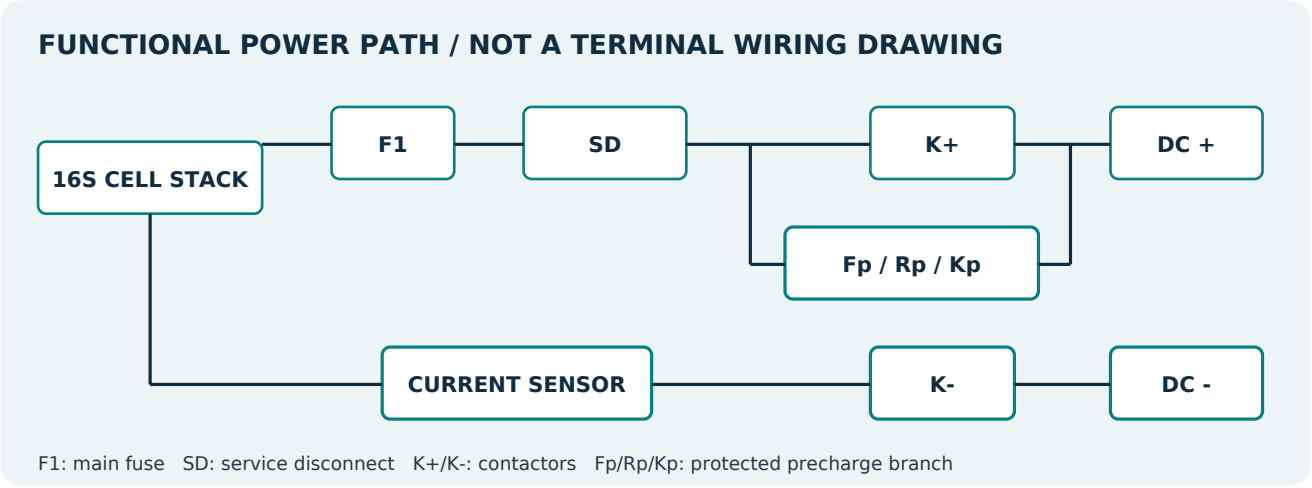
Make an industrial electrical battery

A step-by-step LFP pack build, with a separate route for manufacturing lithium-ion cells.



Worked example: 16S1P lithium iron phosphate (LFP), 100 Ah, 51.2 V nominal, 5.12 kWh nominal. Indoor stationary DC module; 50 A continuous discharge is a design target, subject to thermal and protection validation.

Inside: parts and tooling, 18 pack-build operations, 12 cell-manufacturing operations, inspection gates, test plans, troubleshooting and a printable production record.

Use this as an engineering master guide. The selected cell, BMS, charger, enclosure and protective devices must be converted into a signed, model-specific build package before production.

Prepared 4 November 2026. Intended for a competent industrial team. Drawings and calculations are explanatory; this guide is not a certified battery design or a released factory work instruction.

Define the battery and its controlled settings

This guide covers two routes: **Route A** assembles a battery from qualified, already-formed cells; **Route B** manufactures cells from purchased battery-grade materials. Start Route A at Step 1. Start Route B at C01, then use qualified output in Route A. Cell manufacturing and pack assembly are distinct capabilities offered by industrial development facilities such as UKBIC. [1]

Worked design item	Reference value / required decision
Chemistry and topology	LFP/graphite; 16 cells in series, one cell per position (16S1P). Choose a single exact cell part number and revision.
Nominal rating	16 × 3.2 V = 51.2 V; series capacity remains 100 Ah; nominal energy = 5.12 kWh.
Proposed operating targets	50 A continuous discharge; 20 A normal charge. These targets require confirmation against every component and the duty cycle.
Usable energy example	At an 80% SOC window, approximately 4.10 kWh before losses and ageing. Actual delivered Wh must be measured.
Application boundary	One indoor stationary module. Parallel packs, series-connected modules, vehicle use, inverter and building installation need additional design work.

Complete the settings register before build

Enter the supplier-supported cell operating range, normal charge target, protection thresholds/delays/recovery, charge/discharge temperature maps, terminal specification, restraint requirements, storage SOC and shipping conditions. Add approved torque values, insulation-test settings and acceptance limits. Every entry needs a source document and an approver.

Separate **normal operating limits** from **protective trip limits**. The charger should finish normally before a protection trip is needed. Do not infer full-charge voltage from the nominal 3.2 V label. A pack-voltage reading cannot show whether one series cell is outside its limit.

The cell is selected by controlled documentation, not a generic chemistry label. Replacement suppliers, terminal designs or cell revisions trigger engineering review. Values left unspecified in this guide must be resolved in that review, rather than filled with a plausible guess.

Prepare parts, equipment and documents

Item	Required specification / evidence
16 matched cells	Same qualified model/revision; traceable serials; supplier handling, electrical, thermal and restraint specifications.
16S-capable BMS	Correct cell chemistry/range; compatible sensor inputs, protection outputs, balancing and charger interface; controlled firmware.
Power protection	DC main fuse, disconnect, switching/contactors and protected precharge if required. Ratings and interruption duties established by calculation/testing.
Current path	Qualified busbars, terminals, cables, lugs and keyed output connector; insulation, strain relief and compatible joint materials.
Mechanical / thermal	Enclosure, cell restraints, insulating barriers, terminal covers, cooling path and any engineered venting provisions.
Control / auxiliary	Temperature sensors, protected sense harnesses, auxiliary supply and wiring, charger/load communications and defined safe-state behaviour.
Test / release	Model-specific schematic, assembly drawings, BOM, settings file, risk assessment, test plan and serialised build traveller.

Minimum pack-build tooling

Use calibrated voltage measurement, an appropriate four-wire joint-resistance method, torque-controlled tools and qualified crimp tooling. Provide protected/insulated hand tools, nonconductive covers and fixtures, a cell/pack cycler or matched charger and electronic load, temperature logging, and the design-approved insulation/continuity test setup. Check voltage, current, energy and fault ratings of the test equipment.

Prepare the station and team

Separate receiving/quarantine, mechanical assembly, electrical assembly and energised test. Prevent loose conductive objects and uncontrolled metal debris. Use the approved electrical risk assessment, protective equipment and emergency arrangements. External isolation does not remove stored energy inside cells; cover exposed terminals and minimise the accessible live area throughout assembly.

For cell production, add contained powder handling, controlled clean/dry environments, coating/assembly tools, solvent/exhaust systems and formation facilities. COSHH and DSEAR assessments must reflect actual substances and processes, not the word “battery” alone. [9,10]

Steps 1-3 / Receive and qualify cells

01 / Open the job and verify material identity

Do: Create a pack serial number and traveller. Confirm the released drawing/BOM/settings revisions. Inspect cell shipping condition and identifiers; photograph damage and isolate suspect material. Match the delivered cell to the controlled supplier documentation.

Verify / record: Record supplier, order, cell model/revision, all 16 serials, arrival date and receiving result.

Hold if: Any cell leaks, swells, has damaged terminals/casing or lacks traceable identity; the selected cell revision is not approved.

02 / Measure and characterise the cell lot

Do: At the approved rest condition and temperature, record each cell voltage, dimensions and relevant resistance/impedance measurement. Perform the specified sample or 100% capacity test using the supplier-approved window and rates. Keep the same method and instruments across comparisons.

Verify / record: Retain raw data, measurement conditions, equipment IDs and cell-specific accept/reject results.

Hold if: Abnormal self-discharge, unexplained outliers or a failed acceptance limit. A normal open-circuit voltage is insufficient evidence of cell health.

03 / Match and condition the selected set

Do: Group qualified cells using the approved capacity, resistance and self-discharge criteria. Bring them to the specified assembly SOC with individually controlled, current-limited equipment. Use the approved equalisation method; do not improvise by directly paralleling cells to equalise them.

Verify / record: Record conditioning current, terminal limits, temperature, final condition, rest time and matching result.

Hold if: Conditioning limits are exceeded or the group cannot meet matching criteria without hiding a weak cell.

LFP has a relatively flat mid-SOC voltage curve, so similar resting voltages alone do not demonstrate similar SOC or capacity. Set the matching criteria from pack tests and the intended duty cycle. Independent, consistent preparation and evaluation are central to useful battery comparisons. [17]

Steps 4-6 / Build the mechanical assembly

04 / Prepare enclosure, barriers and fixtures

Do: Check the enclosure against its drawing. Remove debris; inspect edges, insulation, spacers, supports, connector mounts and cooling interfaces. Verify that conductive fasteners cannot contact cell terminals or abrade insulation. Identify intended protective-earth/bonding points where the design requires them.

Verify / record: Drawing revision; dimensions; insulation materials; inspection of clearances, vents and mounting features.

Hold if: Missing barriers, sharp edges, unapproved materials or obstructed thermal/vent paths.

05 / Place and restrain the cells

Do: Install cells in the approved order and orientation using a positioning fixture. Fit separators and restraint components. Apply only the cell supplier’s qualified restraint/compression arrangement, accounting for expansion, temperature and tolerances. Keep vents unobstructed and protect the cell cases as specified.

Verify / record: Cell-to-position map; polarity inspection; restraint setting/method; mechanical inspector sign-off.

Hold if: A cell is reversed, damaged, forced into place or compressed using an unverified torque-to-force assumption.

06 / Fit the series interconnections

Do: Keep external circuits disconnected and terminal exposure controlled. Install the 15 inter-cell links from the approved series drawing, covering completed areas as work progresses. Use specified busbar materials, joint stack-up and joining process. For bolted cells use the exact terminal torque; for welded designs use the qualified weld schedule.

Verify / record: Joint IDs; torque/process trace and tool calibration; witness inspection; specified joint-resistance checks.

Hold if: Wrong polarity, terminal rotation/damage, incorrect hardware, missing evidence or abnormal resistance. Never infer torque from thread size alone.

Verification gate: with the approved protected measurement method, check each series section and cumulative polarity against the drawing. The final stack voltage must equal the sum of the measured cell voltages within measurement uncertainty. This check does not replace the sense-harness inspection in Step 10.

Steps 7-9 / Install power protection and sensors

07 / Install the protected main power path

Do: Fit main fuse, disconnect, switching, busbars and output connector to the drawing. Keep the disconnect open and output unavailable during assembly. Locate protection to limit unprotected conductor length. Confirm current, DC voltage and interrupt/break ratings against the prospective fault duty; include charger and load interfaces.

Verify / record: Device part numbers; fuse coordination study; cable/busbar ratings; assembly/joint records and conductor routing.

Hold if: AC-only ratings are substituted, fault-interruption capability is unknown or one unprotected branch remains.

08 / Install precharge where required

Do: For a capacitive load, fit the approved protected resistor/switch branch and bus-voltage feedback. Verify the state sequence, timeout and no-load condition during precharge. The load must remain inhibited until successful completion. Driver hardware and suppression must match the selected contactors. [6]

Verify / record: Precharge design calculation, pulse-energy rating, repetitive-duty margin, fault protection and feedback map.

Hold if: The connected load capacitance is unknown, timeout is unverified or failure can leave a sustained load on the resistor.

09 / Fit current and temperature measurement

Do: Install the selected shunt or isolated current sensor in the correct direction, ensuring no power path bypasses measurement. Fit and secure temperature sensors at design-validated representative/hot locations with the approved electrical isolation and thermal contact.

Verify / record: Sensor IDs, polarity, range, placement and lead protection; wiring continuity and strain-relief checks.

Hold if: Any unintended current bypass, sensor location change, exposed conductive sensor or incompatible input type.

Worked precharge arithmetic only: for a hypothetical 58.4 V maximum bus, 5 mF load and 100 ohm resistor, initial current is 0.584 A, time constant 0.50 s, and 95% voltage takes about 1.50 s. Resistor energy approaches 8.53 J and initial power is 34.1 W. Actual selection needs tolerances, switching transients, repeated starts and fault-duty protection; these example values are not a BOM.

Steps 10-12 / Connect and configure the BMS

10 / Build and verify the sense harness

Do: Use the released terminal-to-pin schedule. Fit approved cell-tap protection, routing, insulation and strain relief. Verify each pin and adjacent-cell voltage before mating any connector to the BMS, using the manufacturer’s procedure and protected probing method. Grouping and reference pins vary by BMS. [5]

Verify / record: Independent pin-to-cell check, connector inspection, adjacent/cumulative voltage record and technician signature.

Hold if: Any missing, reversed, duplicated or unexplained voltage reading. Do not use a generic 17-wire assumption as a pinout.

11 / Connect control and auxiliary circuits

Do: Follow the exact BMS connection sequence for power, sensors, contactor drivers, auxiliary inputs and communications. Verify protected auxiliary supply, shutdown path, charger inhibit and load permission. Keep main switching in the safe state. Do not alter terminal or sense wiring with the BMS attached contrary to its manual.

Verify / record: Hardware revision; auxiliary voltage; input/output map; communications termination and control-circuit inspection.

Hold if: The BMS hardware is incompatible, the power-up sequence is unknown or an energisation output can become active unexpectedly.

12 / Load the approved configuration

Do: Set cell count, chemistry, current-sensor calibration, charge/discharge maps, temperature limits, balancing, trips, recovery, communications and switch sequencing. Read back the loaded file, compare it with the master and lock change access according to the quality procedure.

Verify / record: Firmware version, configuration ID/checksum, readback comparison, approver and date.

Hold if: Default settings remain unreviewed, normal charging depends on hitting a protection trip, or one cell can violate its limit while pack voltage appears acceptable.

Balancing: select the balancing method and operating region from the cell/BMS design. Balancing does not repair capacity loss, excessive self-discharge or a damaged cell. Validate that it works at the intended charge profile and that it does not create an uncontrolled heating condition.

Steps 13-15 / Inspect and perform initial checks

13 / Complete the physical inspection

Do: Verify the complete cell map, conductor route, joint records, harness clearances, insulating covers, cooling, restraint and connector retention. Count tools/loose hardware. Photograph the approved internal layout before fitting the enclosure cover. Close and seal only to the specified process.

Verify / record: Mechanical inspection report, internal photos, cover/seal records, label and accessible-isolation check.

Hold if: Any unresolved build deviation or a physical change not captured by the drawing/BOM.

14 / Run the unpowered electrical checks

Do: Keep main outputs isolated. Confirm output polarity, sense mapping, continuity and absence of unintended case-to-live connections. Perform bonding/insulation tests only with the approved test points, voltage, duration and electronics-disconnection method. A generic megohmmeter or hipot setting can damage attached electronics.

Verify / record: Test setup and calibration; voltage/polarity results; design-defined insulation and bonding measurements.

Hold if: A result is outside the released criterion, electronics protection is unclear or a test setup differs from qualification.

15 / Power the controls and validate readings

Do: Use the approved protected auxiliary supply with power outputs inhibited. Compare BMS cell/pack voltages and temperatures with calibrated reference measurements. Verify current zero and sign. Exercise permissions and switching logic using a safe simulator/fixture before attaching a live load.

Verify / record: Measurement errors, sensor plausibility, fault flags, auxiliary consumption and controlled shutdown behaviour.

Hold if: Readings are inconsistent, faults are suppressed to obtain a pass or a communications failure leaves unsafe permissions.

Gate before first energy transfer: a second authorised person reviews Steps 1-15, confirms the charger/load profiles and test limits, checks the thermal logging and stop conditions, and signs permission for controlled commissioning. Cells remain energetic even when both contactors are open.

Steps 16-18 / Commission, characterise and release

16 / Perform controlled first power-up

Do: Connect only the approved, isolated test fixture with charger/load disabled. Run the switching/precharge sequence and confirm feedback, bus rise and timeout response. Begin at an engineer-approved low current; 5 A is an illustrative 0.05C trial for the 100 Ah reference, only if all components permit it. Log every cell, current and temperature.

Verify / record: Current direction, permission response, contactor state, bus behaviour, temperature trend and emergency stop.

Hold if: Unexpected inrush, voltage excursion, local heating, odour, swelling, noise or an unexplained protection event.

17 / Measure capacity, energy and thermal performance

Do: Run the released charge/discharge/rest profile. A proposed reference characterisation is 20 A (0.2C), 25 ±2°C, within the selected cell's approved voltage limits. Integrate Ah and Wh at pack terminals. Increase to the 50 A continuous design target only through the approved staged thermal test.

Verify / record: Raw time series; per-cell minima/maxima; capacity/energy; temperatures; efficiency where measured; pass against the declared rating under defined conditions.

Hold if: The limiting cell ends discharge early, joint temperature is abnormal or declared performance cannot be achieved without exceeding limits.

18 / Close the traveller and authorise release

Do: Complete the approved production screens and review results. Resolve deviations; retain the as-built configuration, cell map and test data. Apply accurate ratings/traceability labels and the model-specific storage/shipping condition. Issue installation, operation and service documentation.

Verify / record: Quality signature, release date, software/hardware revision, applicable conformity/transport records and customer handover pack.

Hold if: Any safety nonconformance, missing test record, unapproved rework or a claim outside the qualified operating envelope.

Routine commissioning is not formation: purchased cells should already have been formed and qualified by their manufacturer. Do not attempt to apply a raw-cell formation recipe to a finished purchased-cell battery. Design validation and production acceptance remain separate programmes.

Set acceptance tests before building

The following is a proposed verification matrix. Engineering must assign the numeric acceptance value, method, uncertainty and sampling frequency for the actual model. No universal fuse size, terminal torque, cell mismatch or insulation threshold follows from “industrial grade”.

Test	Record and acceptance basis	Typical stage
Cell and BOM identity	Correct approved part/revision; full lot/serial genealogy.	Every unit / incoming
Joints and restraints	Approved torque/weld/restraint settings; measured process evidence.	Every unit + sample correlation
Electrical mapping	All pin/polarity checks pass; BMS readings agree within approved uncertainty.	Every unit
Insulation / bonding	Released test method and pass limits; intended case/earth architecture.	Every unit as specified
Capacity / energy	At least the declared rating under its stated protocol; no protection limit exceeded.	Qualification; production frequency by plan
Thermal performance	Cell, joint, cable and device temperatures within validated operating limits.	Qualification + production screens
Protection / faults	Correct inhibit/shutdown/recovery under simulated sensor, communication and driver faults.	Qualification + approved EOL checks
Ageing / durability	Cycle/calendar data, vibration/ingress where relevant, and applicable abuse/propagation testing.	Design qualification
Transport / product safety	Reports for the released design and intended market/application.	Before relevant shipment / sale

Test faults with fixtures, not improvised abuse

Use cell simulators, sensor substitution and hardware-in-the-loop fixtures to test trip thresholds, timing and recovery. Do not create real overcharge, short-circuit or thermal-abuse events on the production bench to prove a software setting. Specialist facilities conduct required destructive tests under a defined plan.

A commissioning pass cannot substantiate a long-life warranty. Continue calendar/cycle testing across representative lots and monitor field configurations. Berkeley research shows that charging risk depends on conditions such as temperature, SOC and design, so a current limit alone is incomplete. [15]

Cell production / C01-C04: formulate and coat

Route B uses a qualified industrial laboratory/pilot line. Purchase battery-grade LFP, graphite, conductive additives, binders, foils, separator and electrolyte from approved suppliers. The sequence follows the conventional electrode/assembly/finishing structure; the control plan below is a proposed engineering synthesis. [1,2,18]

C01 / Freeze the cell design and raw-material specifications

Select a single format and chemistry combination. Specify capacity, loading, geometry, operating window, electrode capacity balance, separator, electrolyte, housing and current paths. Qualify incoming materials for the properties that affect performance, contamination and moisture.

Release record: Approved design/BOM; supplier certificates verified by sampling; retained material samples; design risk analysis.

C02 / Prepare the anode and cathode mixtures separately

Develop the approved addition order, dispersion energy, solids content, temperature, hold time and degassing. Evaluate rheology over relevant shear conditions and correlate it with coating behaviour. Binder/solvent routes depend on chemical compatibility; water is not a universal substitute. [4]

Release record: Recipe revision, material weights/lots, mixing history, solids/rheology result and maximum-use-time status.

C03 / Coat the current collectors

Apply the qualified cathode mixture to the specified aluminium foil and the qualified graphite anode mixture to the specified copper foil. Control loading, coating geometry, two-sided registration and edge features. Map defects along and across the web, rather than accepting the roll by average mass alone.

Release record: Roll ID, web position map, loading/thickness, defect locations, process settings and disposition.

C04 / Dry the coatings

Use the qualified staged drying process and appropriate extraction/recovery. Verify residual solvent/moisture, adhesion, cracks and microstructure-related properties. Set limits from the chosen materials and downstream performance. Drying temperature alone does not establish that the electrode is ready.

Release record: Actual thermal/exposure history, residual measurements, adhesion/conductivity checks and released roll regions.

Cell production / C05-C08: finish and assemble

C05 / Calender the electrodes

Compress the coating using the validated roll gap/load, temperature and speed. Measure thickness, density/porosity and spring-back after the defined rest. Confirm the selected density still permits wetting, ionic transport and rate performance; maximum compression is not the objective.

Release record: Roll settings, dimensional map, retained electrode samples and electrochemical correlation.

C06 / Slit or cut and final-dry

Cut the electrode geometry with controlled burrs, particles and tool/laser condition. Remove contamination using the approved method. Final-dry as required and transfer/store under a validated clean/dry protocol with controlled exposure time. Keep electrode-position identity into assembly.

Release record: Geometry, edge/particle inspection, material moisture, storage/exposure history and cut-part identity.

C07 / Stack or wind with the separator

Assemble the electrodes and separator to the format-specific drawing. Verify polarity, electrode overlap, separator coverage, tension/position and absence of folds or damage. Use qualified inspection that has been challenged with relevant known defects.

Release record: Assembly geometry and inspection data; component lots; equipment/fixture revision and operator.

C08 / Join current paths and package

Join tabs/collectors using the qualified process and fit the approved insulators, housing and closure components. Validate joint strength/resistance and control joining debris. Inspect the internal assembly before it becomes inaccessible. Keep the format-specific filling/venting provisions available.

Release record: Joint process trace and sample tests; enclosure/insulation checks; cell serial number linked to electrode positions.

Facility control: define particle and humidity conditions by station and chemistry, then qualify the transfer route. UKBIC publishes distinct controlled environments for electrode production, assembly and electrolyte filling; those examples show the need for environmental control, not one universal setpoint for all cells. [1]

Cell production / C09-C12: fill and activate

C09 / Fill and verify electrolyte wetting

Use the qualified closed filling operation with controlled electrolyte identity, amount, exposure and pressure/dwell sequence. Establish wetting evidence appropriate to the format; a correct fill mass alone does not prove that every active region is wetted. Quarantine leaks, gas anomalies or unexplained nonuniformity.

Release record: Electrolyte lot, fill mass, filling history, wetting checks, moisture/exposure records and inspection result.

C10 / Seal and check containment

Apply the qualified format-specific sealing/leak-test sequence. Some designs require provisional sealing and later formation-related degassing before final closure; document that sequence explicitly. Validate leak-detection capability against known references and inspect seal integrity.

Release record: Seal settings, leak-test setup/result, closure revision and rework disposition.

C11 / Run the validated formation programme

Apply the approved charge/discharge/rest, temperature and mechanical conditions with independent channel limits and emergency response. Manage any format-specific gas removal and final sealing. Formation establishes protective interphases and must be co-developed with materials and design; there is no universal LFP waveform. [3,16]

Release record: Time-series voltage/current/temperature, initial efficiency, gas/swelling observations, channel calibration and final-seal record.

C12 / Age, screen and release the cell lot

Store and screen under the qualified conditions for capacity, resistance, self-discharge, dimensions and other relevant indicators. Investigate outliers and perform the defined sample teardown, ageing and safety tests. Release only cells within the validated design/process window, then return to pack Step 1.

Release record: Cell-level acceptance results, raw data, lot distributions, deviation review and quality release signature.

Development gate: demonstrate reproducibility on independent lots and representative full-format cells before increasing line speed. Keep the measured baseline when changing electrode formulation, drying, formation or screening. A small laboratory cell result does not qualify a large production cell. [17]

Run the factory and resolve defects

Observed issue	Investigate	Disposition
Pack resistance / hot joint	Joint stack-up, terminal condition, joining process, sensor placement and measurement method.	Stop; isolate affected units; use qualified repair and full retest.
One cell reaches a limit early	Capacity/SOC mismatch, self-discharge, tap wiring, local thermal condition or degraded cell.	Quarantine; diagnose before any replacement or reconditioning.
Precharge fails	Bus capacitance, load inhibit, resistor/switch, bus measurement, auxiliary power and timing.	Keep output inhibited; do not repeatedly retry into an unknown fault.
BMS voltage mismatch	Pin mapping, open/shorted sense lead, connection resistance, input compatibility or calibration.	Disconnect/rework only under the exact BMS safe sequence; reverify harness.
Cell-lot self-discharge	Contamination, moisture, assembly damage, sealing or formation variation.	Contain lot; retain samples; investigate upstream genealogy.
Electrode loading drift	Mixture ageing/rheology, pump/die, web condition and metrology.	Segregate mapped web region; correct cause and validate restart.

Use a controlled stop-and-restart process

Stop production at the defined trigger. Identify the last known good state; contain affected lots and downstream units. Record the symptom and evidence before changing settings. Engineering defines root-cause work, authorised rework and restart checks; quality confirms the contained population and final disposition.

Track useful manufacturing measures

Track first-pass yield, saleable yield, defects by station, rework, test escapes, thermal/formation incidents and field returns. Keep availability/rate losses separate from quality rejects. Measure energy and material consumption per saleable kWh, not only gross capacity processed. Preserve calibration, maintenance and recipe histories.

Retain an auditable material-to-field genealogy. A replacement cell or BMS setting change creates a new as-maintained configuration and may require revalidation; replacing a serial number in a spreadsheet is insufficient configuration control.

Production traveller and commissioning record

Printable record template. Complete the linked detailed worksheets and raw datasets for the selected model. A blank line is not a pass; missing evidence holds the unit.

Record field	Entry / linked evidence
Job / pack serial / date	_____
Drawing / BOM / firmware / settings revisions	_____
Cell model / revision / supplier lot	_____
Cell serial map / incoming report	_____
Matching / conditioning dataset	_____
Terminal joint / restraint records	_____
Fuse / contactor / precharge part IDs	_____
Sense-harness check / inspector	_____
Physical / insulation / polarity checks	_____
Control calibration / permissions / faults	_____
First energisation / precharge result	_____
Capacity / energy / thermal dataset	_____
Nonconformances / authorised rework	_____
Release decision / quality signature	_____

Minimum data to retain

Store timestamped pack/cell voltage, current, temperature and switch/fault states at a rate suitable for the test. Record instrument uncertainty, profile revision, sampling interval and ambient conditions. Link photographs, join/torque traces, configuration files and the authorised pass criteria to the pack serial number.

Set storage and access rules before production. Keep raw results when recalculating capacity or updating analysis software; derived pass/fail labels alone cannot support later investigation. This template is a static printable page, not an interactive or digitally signed form.

Qualify, document and hand over the product

Separate three approvals

Design qualification: demonstrate that the exact battery configuration meets application performance and safety requirements. **Process qualification:** show that the controlled line repeatedly makes that configuration. **Unit release:** confirm that an individual battery passed the specified manufacturing and acceptance checks. None substitutes for the other two.

Requirement	Action before the relevant sale / shipment
Industrial battery safety [7]	Agree IEC 62619 applicability and the relevant product-specific standards with the test laboratory. Road vehicles and other special applications may need different requirements.
Transport [8]	Determine UN 38.3 test and test-summary obligations, design-change implications, packaging and mode-specific dangerous-goods rules with specialists.
GB / NI markets [11,13]	Create a jurisdiction-specific compliance matrix. Current OPSS guidance distinguishes GB law from the EU Batteries Regulation applying in Northern Ireland.
Producer responsibility [12]	Determine registration, chemistry/tonnage reporting, take-back and approved waste routes for the correct battery category.
NI / EU passport planning [13]	Prepare the required data for industrial batteries above 2 kWh placed on relevant markets from 18 February 2027. Confirm full scope and any related phased duties.

Issue a complete handover pack

Provide accurate rated voltage/capacity/energy and operating limits, compatible charger/load interfaces, installation constraints, handling/storage instructions, maintenance, service isolation and fault response. Include serial identification, warranty terms supported by evidence, transport information and end-of-life route. Define who can modify settings or replace cells.

Close the remaining engineering inputs

Before releasing an executable work instruction, select the exact cell, BMS, charger and protection hardware; complete every setting, torque, restraint, wiring, test and acceptance value; qualify the integrated design; and obtain the responsible engineering and quality signatures. UKBIC describes gated development support while the customer remains its product design authority. [14]

Sources and use of this guide

Sources retrieved or verified on 4 November 2026. This guide combines official process descriptions, manufacturer documentation, research and regulatory guidance. The numbered operations, traveller and worked electrical/precharge examples are original engineering synthesis, not a validated supplier build procedure.

[1] UKBIC. Cell Development. Open source
Electrode, assembly and finishing capabilities; facility parameters are not universal process limits.

[2] VDMA / PEM RWTH Aachen. Production Process of a Lithium-Ion Battery Cell, fifth edition, 2026. Open source
Process-chain reference; not a qualified LFP recipe.

[3] Energy & Environmental Science / RSC. Lithium-ion battery cell formation: status and future directions towards a knowledge-based process design. Open source
Formation review and dependence on cell/material design.

[4] Energy Advances / RSC. Battery electrode slurry rheology and its impact on manufacturing. Open source
Rheology, formulation and process-measurement review.

[5] Ewert Energy Systems / Orion BMS. Orion BMS 2 Wiring & Installation Manual. Open source
Manufacturer-specific wiring checks; follow documentation for the actual selected hardware.

[6] Ewert Energy Systems / Orion BMS. Precharge Settings. Open source
Manufacturer example of precharge control. Circuit and timing remain application-specific.

[7] IEC. IEC 62619:2022. Open source
Official industrial lithium-battery safety scope; paid full standard not reviewed.

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[9] HSE. How to carry out a COSHH risk assessment. Open source
Official workplace substance-risk assessment guidance.

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[12] GOV.UK. Waste batteries: producer responsibility. Open source
Registration/reporting and take-back duties.

[13] OPSS / GOV.UK. EU Batteries Regulations in Northern Ireland. Open source
NI framework, phased duties and battery-passport scope.

[14] UKBIC. How We Operate. Open source
Gated scale-up support and customer design responsibility.

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Research on formation/ageing; not a transferable production waveform.

[17] Communications Materials. Best practices in lithium battery cell preparation and evaluation. Open source
Consistent preparation and evaluation; public summary used.

[18] Fraunhofer ISE. Wet and Dry Electrode Manufacturing and Thin-Film Technology. Open source
Electrode processing infrastructure and research context.

Document status and limitations

GB-LFP-001 / v1.0 / Engineering master guide. No particular cell or BMS is selected, no hardware was built or tested, and no certification is claimed. Generic material and process descriptions do not establish a manufacturing operating window. The exact model requires controlled drawings, supplier documentation, risk assessment, validated settings and acceptance tests before production.